

Q1

Q2

Q3

Q4

Q5

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University of Canterbury

Term Test 2024

Prescription Number(s): **ENEL445**

Paper Title: **Applied Engineering Optimisation**

Time allowed: **1 hour 30 minutes**

Number of pages: **19**

Calculators **ARE** permitted.

ALL Questions ARE COMPULSORY.

(There are 5 questions in total.

The maximum total score is 100 marks.)

Family name (in block letters):

Given name:

Student number:

INSTRUCTIONS

This is an closed-book test (apart from the A4 cheat sheet). Please write you answers in the space provided. There is extra space at the end of the test should you need it. Show your working as you complete the test, so that you can more easily receive partial credit. Make a note indicating when your work is continued on an extra page.

Q1. Basics**(a) [6 marks]**

Let $\mathbf{b}$ and $\mathbf{x}$ be column vectors of size $n \times 1$. Show that the vector derivative of $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$ with respect to $\mathbf{x}$ is $\mathbf{b}$.

(b) [2 marks]

Write down the Frobenious norm of a matrix $\mathbf{A}$ in terms of $\mathbf{A}$, $\mathbf{A}^T$, and the trace operator.

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Write down the Taylor series expansion around the point $\mathbf{c}$ for a multivariate function, $f(\mathbf{x})$, up to and including the second order term.

Write down the definition of the Hessian and discuss the role of the Hessian in the context of optimisation, include the term “positive semi-definite” in your answer. Feel free to draw pictures!

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Q2. Unconstrained Optimization (1)**(a) [2 marks]**

Put $\mathbf{x} = \mathbf{c} + \mu\mathbf{p}$ into your answer to Q1 (c), and keep the first-order term only to derive the first-order Taylor-Series approximation of $f(\mathbf{x})$ along the direction $\mathbf{p}$.

(b) [4 marks]

Assume that the direction vector $\mathbf{p}$ has unit length (i.e., $\|\mathbf{p}\|_2 = 1$).

Use your answer to Q2 (a) to show that at the point $\mathbf{c}$, the negated gradient $-\nabla f(\mathbf{c})$ gives the direction of the fastest function **decrease**.

(c) [4 marks]

Explain why we must have $\nabla f(\mathbf{c})^T \mathbf{p} = 0$ for any direction $\mathbf{p}$, if $\mathbf{c}$ is a local minimum of $f(\mathbf{x})$.

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(d) [5 marks]

Calculate, using the Chain Rule, the partial derivative of $f(\mathbf{c} - \mu \nabla f(\mathbf{c}))$ with respect to the step size μ . That is, find $\frac{\partial f(\mathbf{c} - \mu \nabla f(\mathbf{c}))}{\partial \mu}$.

(e) [5 marks]

Suppose when $\mu = \mu^*$, we have $\frac{\partial f(\mathbf{c} - \mu^* \nabla f(\mathbf{c}))}{\partial \mu} = 0$.

Use your result in Q2 (d) to show that the gradient of $f(\mathbf{x})$ at the point $\mathbf{c} - \mu^* \nabla f(\mathbf{c})$ is orthogonal to the gradient of $f(\mathbf{x})$ at the point $\mathbf{c}$.

That is, prove that $\nabla f(\mathbf{c} - \mu^* \nabla f(\mathbf{c}))^T \nabla f(\mathbf{c}) = 0$.

Q3. Unconstrained Optimization (2)**(a) [8 marks]**

In unconstrained optimization, the generic form for updating the design variable $\mathbf{x}$ of the objective function $f(\mathbf{x})$ is $\mathbf{x}_k = \mathbf{x}_{k-1} + \alpha_k \mathbf{p}_k$. Here, $\mathbf{x}_k$ is the solution obtained in iteration k , α_k is the step size and $\mathbf{p}_k$ is the search direction.

Show that this generic update form can actually be derived by solving another unconstrained optimization problem:

$$\min_{\mathbf{x}_k} h(\mathbf{x}_k) = \frac{1}{2\alpha_k} (\mathbf{x}_k - \mathbf{x}_{k-1})^T (\mathbf{x}_k - \mathbf{x}_{k-1}) - (\mathbf{x}_k - \mathbf{x}_{k-1})^T \mathbf{p}_k.$$

Hint: First find the partial derivative of $h(\mathbf{x}_k)$ with respect to $\mathbf{x}_k$. Then, setting the result to zero would yield the desired answer.

(b) [4 marks]

What insights can you obtain from examining $h(\mathbf{x}_k)$? Explain your answer.

TURN OVER

(c) [8 marks]

We are trying to solve a weighted nonlinear least squares (NLS) problem, whose objective function is

$$f(\mathbf{x}) = \frac{1}{2}(\mathbf{b} - \mathbf{g}(\mathbf{x}))^T \mathbf{W}(\mathbf{b} - \mathbf{g}(\mathbf{x})).$$

Here, $\mathbf{W}$ is a positive definite weighting matrix. $\mathbf{b}$ is the measurement vector and $\mathbf{g}(\mathbf{x})$ is a nonlinear **vector** function of $\mathbf{x}$.

We are going to update the solution iteratively. At iteration k , we first approximate $\mathbf{g}(\mathbf{x})$ using $\mathbf{g}(\mathbf{x}) \approx \mathbf{g}(\mathbf{x}_{k-1}) + \mathbf{J}_{k-1}(\mathbf{x} - \mathbf{x}_{k-1})$, where $\mathbf{J}_{k-1}$ is the Jacobian matrix.

Derive the desired update rule by putting the above approximation into $f(\mathbf{x})$ and finding $\mathbf{x}_k$ that minimizes the approximated objective function.

Q4. Constrained optimisation & algorithms**(a) [16 marks]**

Consider the problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 - x_2 \\ \text{subject to} \quad & x_2 \leq -1 \\ & x_1 - x_2 \leq 0 \end{aligned}$$

i) State the Lagrangian function, including slack variables [2 marks].

ii) Derive the KKT conditions associated with this problem and find the optimal solution [8 marks].

TURN OVER

[illegible]

iii) State the Lagrangian dual problem, identifying the dual function (note: you are NOT required to work out the dual function - although you would not be penalised for doing so) [2 marks].

iv) Explain what *duality gap* is. What is the duality gap in this case? Prove your answer mathematically or otherwise [4 marks].

In a few sentences, explain the concept of *interior penalty*.

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Q5. Optimisation-based Control**(a) [10 marks]**

Consider the system:

$$x_{k+1} = f(x_k, u_k)$$

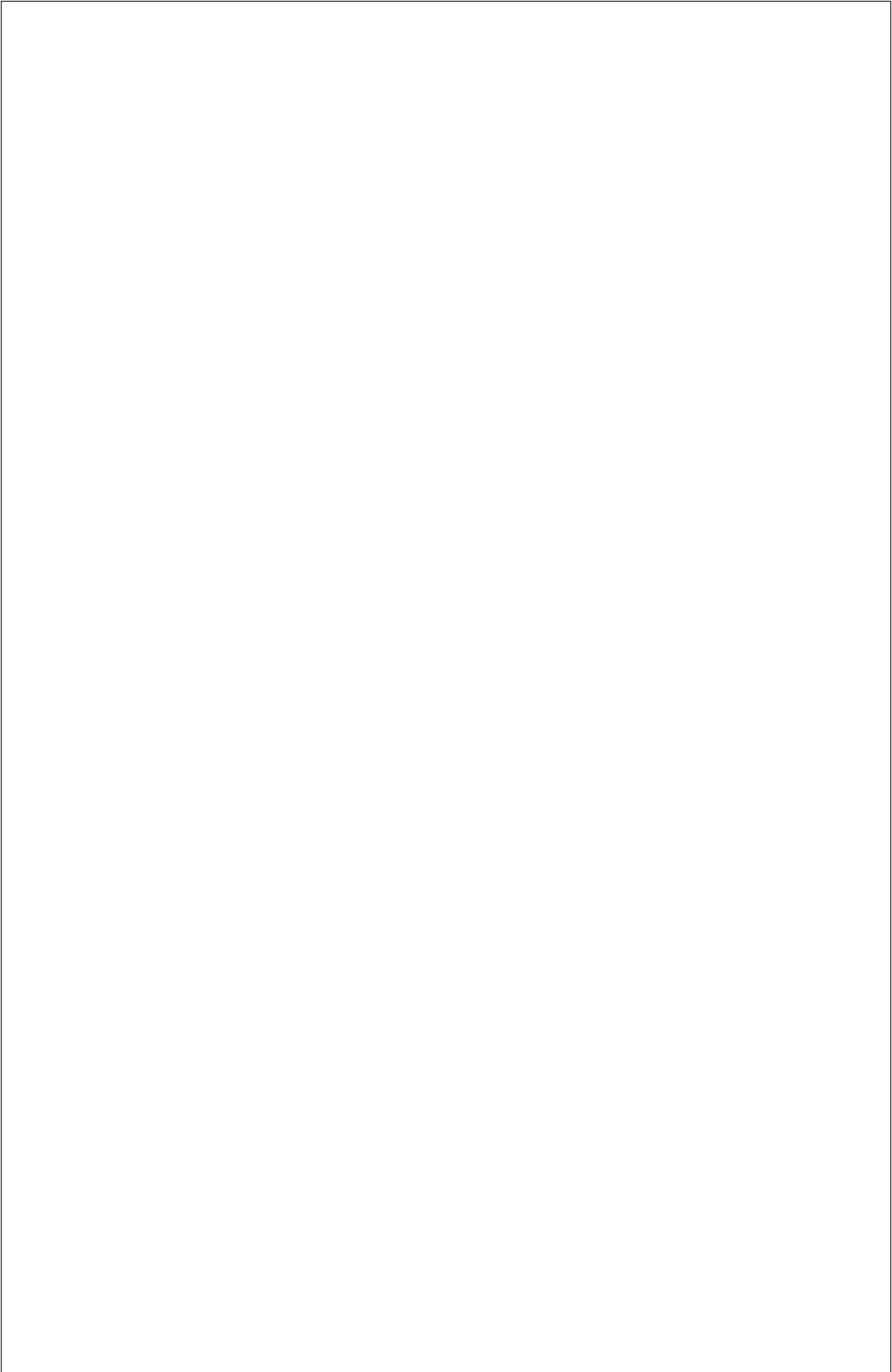
with

	$u = 1$	$u = 2$	$u = 3$
$f(x_k, u_k) =$			
$x = 1$	1	2	3
$x = 2$	1	3	1
$x = 3$	2	1	2

and cost function:

	$u = 1$	$u = 2$	$u = 3$
$c(x_k, u_k) =$			
$x = 1$	1	3	4
$x = 2$	1	2	3
$x = 3$	0	1	2

and terminal cost $J_h(x_h) = 3 - x_h$. Find the optimal solution for two steps (i.e. the optimal u_{h-1} and u_{h-2}) for any initial value of $x_{h-2} \in \{1, 2, 3\}$ recursively using dynamic programming. (Hint: a graphical illustration might be helpful).



TURN OVER

What is Q Learning? Illustrate your answer with pseudo-code describing a Q - Learning algorithm using ϵ -greedy exploration. Explain the main parameters, what they are, and how they work and/or why they are needed.

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[illegible]

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